When Does the Router Matter?
A Two-Factor Law and a Structural Attractor in Graph Load Balancing, Found Through Classical-Physics Heuristics

Carlos López

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Abstract

Load balancing on a graph is usually framed as algorithm design: a better router yields a better outcome. We present evidence for an empirical two-factor law that states when the router matters at all. Treating packets as particles under classical forces, we built independent routing cores—a Newtonian loss-reaction, an Archimedean buoyancy, a Pascalian pressure diffusion, a Hookean spring—that share a search substrate but no mechanism, and used them as instruments on a bench of fifteen topologies (grids, small-world, scale-free, and the real GÉANT and Abilene backbones). Five findings. First, an *attractor*: all congestion-aware rules produce near-identical per-edge load distributions (cosine 0.979, against 0.867 for blind ECMP; Wilcoxon $p = 10^{-4}$ on two metrics)—topology and demand fix most of the outcome. Second, an *interventional law*: scaling link capacities within fixed topologies shows that congestion *causes* mechanism divergence (rank correlation up to 1.0; a squeezed scale-free graph diverges 21×, a relieved backbone converges 8×), bounded by a structural ceiling—the real Abilene, our most congested graph, cannot express divergence its four cycles leave no room for. Third, a *double dissociation*: a structural ratio (tube/sp, computable before deployment) predicts the gains congestion-aware rerouting can deliver (partial $r = +0.59$ in sample, $r = 0.85$ out of sample on eighteen unseen SNDlib backbones, and $r = 0.77$ under those backbones' real measured demand), while congestion predicts how much the mechanism's identity matters (partial $r = +0.92$). Fourth, a *validated minimal core*: gradient plus an anticipatory threshold force beats the bare substrate on fifteen of fifteen topologies in every regime ($p \leq 10^{-4}$), across a broad parameter plateau and for two functional forms—while the reactive scar of our own earlier work does not replicate there, a verdict our re-audit inverted and we report in full. Fifth, a *taxonomy of information channels*: on a non-stationary bench of episodically failing links, the retired temporal term returns to beat every memoryless core including the stationary champions ($p \leq 10^{-3}$), its scar identifying flaky edges with precision 0.69–0.80 where load statistics reach 0.20—threshold owns the nonlinear present, Pascal space, Newton time, and the regime decides which channel pays. The boundary is equally explicit: in saturation no physical core reaches statistical equivalence with the engineered balancers CONGA and DRILL at any margin—engineering matters exactly where the law says it must.

1 Introduction

Network load balancing is a mature engineering discipline. Production systems such as CONGA [2] and DRILL [8] spread traffic across a network using carefully designed mechanisms: per-flowlet rerouting, multi-path candidate enumeration, local queue sampling, and centrally or distributively maintained congestion state. These systems work well, and improving on them is hard.

This paper does not try to improve on them. It asks a different and, we think, more interesting question:

> *How much of what engineered load balancers achieve can be recovered from classical physics alone—from a packet treated as a particle obeying a couple of elementary mechanical laws?*

The motivation is partly methodological. An expert in routing reasons within the conceptual vocabulary of routing; a physicist handed the same problem reasons about gradients, forces, and conservation laws. Occasionally the second framing reaches a familiar destination by an unfamiliar and more economical road, and the comparison is informative regardless of which arrives first. Our interest is therefore not a horse race but a question of *parsimony*: how little mechanism suffices to reach the operating point that elaborate engineering reaches.

1.1 The model, in one paragraph

We treat a packet as a particle descending a potential field defined on the network graph. The particle feels two forces. The first is a *gradient*: it rolls toward lower latency and lower congestion, subject to a hop budget that lets it take a modestly longer, emptier detour rather than forcing every packet onto the single shortest path. The second is a *reaction* in the sense of Newton's third law: when a packet is dropped on a congested edge, it pushes back, depositing a decaying "scar" that steers subsequent packets away from edges that have just failed. That is the entire router—two terms, no central controller, no per-flow state, no precomputed path tables, sixty lines of code. (Sixty lines describe the mechanism, not the operating cost; Section 13.5 prices that difference.)

1.2 What we find

Our central result is deliberately stated as an *equivalence*, not a victory:

1. **Two terms suffice.** The two-term core is the survivor of an ablation: an earlier model carried eight physically-motivated layers (effective mass, a thermostat, a temporal ripple, buoyancy), and measurement showed that only the gradient and the Newton-III scar earn their keep—the rest are inert or mildly harmful (Section 4).

2. **It matches engineered routers.** Tested as a statistical equivalence (TOST, margin 2 percentage points) across fifteen topologies in a flow-level simulator, the core is equivalent to DRILL on twelve of fifteen, and equivalent to CONGA once CONGA is given a fair path budget. It reaches the same operating point CONGA reaches only after enumerating and scoring sixteen to thirty-two candidate paths per flow (Sections 6–7).

3. **A topological metric predicts when.** A quantity computable from the graph alone, the tube/sp ratio (room to manoeuvre among near-shortest paths), predicts where the equivalence holds and where engineered multi-path routing keeps a genuine edge (Pearson $r = 0.78$); the two real WAN backbones, with low tube/sp, are exactly the predicted exceptions (Section 12).

The upshot is that two classical forces, with none of the apparatus of modern load balancers, reach the same place those balancers reach, on the topologies where the geometry leaves room to do so—and that the qualifying clause is itself predictable.

1.3 A note on method

This paper reports its negative results and its own corrections in full. The eight-layer model is presented and then dismantled; an early, larger advantage over CONGA is shown to have been an artifact of an unfair path budget and is retracted (Section 7). We regard this not as throat-clearing but as the substance of the contribution: a claim that physics matches engineering is worth precisely as much as the adversarial scrutiny it has survived, and we would rather report an honest equivalence than an inflated win. Every headline figure here has been checked against the strongest version of its baseline we could build.

One reading note. Sections 3 and 4 present the model as it was originally conceived and reduced—gradient plus a Newton-III loss reaction—because the re-audit that later *inverts* that verdict (Section 5) is itself one of the results. The reader should know the destination in advance: the core this paper ultimately validates is gradient plus an anticipatory threshold force; the two-factor law of Section 10 is the central claim; and the physical cores are best understood as instruments. We preserve the chronology because the inversion, and the machinery that caught it, are part of the contribution.

2 Related Work

Engineered load balancing. Modern multipath load balancers spread traffic to avoid hot spots. ECMP [9] splits flows across equal-cost shortest paths by hashing, with no congestion awareness. CONGA [2] adds that awareness: it enumerates a set of candidate paths and selects among them using in-network congestion feedback, rerouting at flowlet granularity. DRILL [8] pushes the decision to the switch and the microsecond, making per-packet local choices based on queue occupancy with minimal state. These systems define the operating points we measure against. Our aim is not to outperform them—on the topologies where they are strong, they remain strong—but to ask how much of their behaviour is recoverable from a far simpler, physically-motivated rule. Transport-layer multipath—MPTCP [7] and the in-progress multipath extension of QUIC [10]—is orthogonal to this in-network family: endpoints spread load across whole paths end to end, while the balancers above, and our core, choose next hops inside the network.

Physics- and nature-inspired routing. Using physical or biological analogies for routing has a long history: ant-colony optimisation [6], electrical-network and resistance analogies, potential- and field-based forwarding, and physarum-inspired path formation [14]. These approaches typically introduce a mechanism and demonstrate that it works. Our contribution differs in emphasis in two ways. First, we run the analogy in reverse as an ablation: rather than showing that added physics helps, we add a great deal of physics and then measure how much can be removed, arriving at a two-term residue. Second, we frame the outcome as a statistical equivalence against strong engineered baselines, rather than as a demonstration that the physical method is novel or superior.

Scope of comparison. We benchmark against CONGA and DRILL because they are the state-of-the-art in adaptive, congestion-aware load balancing—the exact problem the physical core addresses. Schemes that do not solve that problem are out of scope, not because they are unimportant but because they answer a different question: shortest-path and ECMP appear only as blind references (they ignore congestion by construction), and legacy access or medium-arbitration protocols (token passing, spanning-tree-era schemes) are not load balancers at all. Equivalence with the best congestion-aware balancers is the claim that carries weight; equivalence with a non-adaptive scheme would not.

Backpressure and queue-gradient methods. Backpressure scheduling [13] and its delay-aware refinements [11] route along gradients of per-queue backlog, and are throughput-optimal under stationary arrivals. The connection to the present work is direct: our load term is itself a queue-gradient signal, and the two-term core can be read as a backpressure-like rule augmented with a shortest-path bias and a slow scar memory. The distinction we draw is not mechanistic but evaluative—we ask how close such a rule sits to engineered multipath balancers, and report the answer as an equivalence with a stated margin rather than as a throughput-optimality proof.

Learning-based routers. A parallel line replaces hand-built rules with learned policies, from Q-routing [4] to graph-aware deep reinforcement learning [15]; recent surveys document a growing body of such routers [16]. These trade interpretability for adaptivity. Our contribution is complementary and deliberately opposite in spirit: a fixed, inspectable rule with no training, offered not as a performance ceiling but as a parsimony baseline against which the value added by learning—or by engineered path enumeration—can be measured.

Per-packet and per-flow congestion state. A related family carries congestion state with the traffic itself. ECN [12] marks packets that traversed a congested queue; DCTCP [3] reacts to the fraction marked; ConEx [5] re-exposes that signal for accountability. These mechanisms share with our scar term the idea that a link should carry a memory of recent trouble. The difference is where the memory lives and acts: these feed endpoint congestion control, leaving the in-network forwarding decision stateless, whereas our scar modulates the next routing choice locally, without a round trip. The flaky-link benchmark (Section 13.6) is where that distinction earns its keep.

Equivalence testing. Most networking evaluations test for a difference (one method beats another). We instead test for *equivalence*, using TOST (two one-sided tests), which asks whether two methods are indistinguishable within a stated margin. This is the statistically appropriate tool when the claim is "as good as", not "better than", and it guards against the familiar error of reading a non-significant difference as evidence of equivalence. To our knowledge, equivalence testing is uncommon in the routing-evaluation literature, and its use here is deliberate: the entire thesis is one of equivalence with parsimony.

Applicability conditions. A recurring weakness of method papers is silence about when the method fails. We attach to our result an explicit, topology-only predictor (the tube/sp ratio, Section 12) that says in advance where the equivalence holds and where engineered routing retains an edge. This places the contribution closer to the spirit of complex-systems and network-science work, where structural predictors of dynamical behaviour are themselves the object of study, than to a conventional systems-performance paper.

Conceptually adjacent, though router-free, is the literature on dynamic averaging and load-balancing processes on graphs, e.g. Alistarh et al. [1], where iterative local exchanges converge to equilibria fixed by graph structure. The attractor reported in this paper is the routing-level analogue of that structural determinism, observed empirically across heterogeneous mechanisms rather than proved for a single dynamics.

3 The Two-Term Physical Core

(Presented as built: Section 5 re-audits this core on the modern bench and inverts which term earns its place.)

We model a packet as a classical particle descending a potential field defined on the network graph. The particle is subject to two forces, and only two. We state them first, then justify by ablation (Section 4) that no further mechanism is needed.

3.1 The potential

Each directed traversal of an edge (u, v) carries a cost

$$\phi(u,v) \;=\; \alpha\,\ell(u,v) \;+\; \beta\,\frac{x(u,v)}{c(u,v)} \;+\; \gamma\,s(u,v), \tag{1}$$

where ℓ is edge latency, x/c is the instantaneous load-to-capacity ratio (congestion), and s is a persistent *scar* term defined below. The three weights (α, β, γ) are the only tunable quantities of the model. All three are local: computing ϕ needs no global state beyond the edge itself.

3.2 Force 1: gradient descent

The particle follows the downhill direction of ϕ from source to destination, subject to a hop budget: it may use a path no longer than $\lceil \alpha_{\text{budget}}\, h_{\text{sp}} \rceil$ hops, where h_{sp} is the shortest-path hop count. The minimum-potential path under this budget is found exactly by a dynamic program over (hops used, node), in $O(k\,|E|)$ time with $k = \lceil \alpha_{\text{budget}} h_{\text{sp}} \rceil$. The budget is what lets the particle take a slightly longer, less congested detour instead of forcing every packet onto the single shortest path—the mechanism by which load spreads.

3.3 Force 2: Newton's third law (the scar)

The gradient term reacts to congestion that is *present*. It cannot react to congestion that has just *caused a loss*: once a packet is dropped, the offending edge's instantaneous load may already have fallen. We close this gap with a reaction term inspired by Newton's third law—to every action a packet undergoes, the network responds with an equal and opposite reaction on the packets that follow.

Algorithm 1: EMMET-core routing step (two-term physical router)

Input: graph $G = (V, E)$ with per-edge latency ℓ, load x, capacity c; scar field s; source a, destination z; weights α, β, γ; budget factor α_b

Output: path $a \rightsquigarrow z$

$h_{\mathrm{sp}} \leftarrow$ shortest-path hop count $a \to z$;

$k \leftarrow \lceil \alpha_b\, h_{\mathrm{sp}} \rceil$; `// hop budget`

for *each edge* (u, v) **do**

  $\phi(u, v) \leftarrow \alpha\, \ell(u, v) + \beta\, x(u, v)/c(u, v) + \gamma\, s(u, v)$;

end

$P \leftarrow$ min-$\sum \phi$ path of length $\leq k$ via DP over (hops, node); `//` $O(k|E|)$

return P;

`// On a drop at edge` (u, v) `while forwarding:`

$s(u, v) \leftarrow s(u, v) + 1$; `// Newton-III reaction`

`// Once per step, everywhere:`

$s \leftarrow \rho\, s, \quad \rho = 2^{-1/T_{1/2}}$; `// scars fade`

Concretely, when a packet is dropped on edge (u, v), that edge's scar is incremented, $s(u, v) \leftarrow s(u, v) + 1$. The scar enters the potential (1) with weight γ, so subsequent particles feel a repulsive reaction steering them away from edges that recently failed—information the instantaneous congestion term does not carry.

Scars are not permanent: at each step they decay, $s(u, v) \leftarrow \rho\, s(u, v)$ with $\rho = 2^{-1/T_{1/2}}$ and half-life $T_{1/2}$. A corridor that stops failing reopens on its own, and at rest (no losses) the scar field vanishes and the router reduces to pure gradient descent. The mechanism is self-regulating: it acts only when and where the network is actually breaking. This is the only memory in the model, and—as the ablation will show—the only added mechanism that earns its keep.

3.4 What is deliberately absent

This two-term core is the survivor of a larger model. Earlier versions of EMMET endowed the packet-particle with extra physical attributes: an effective mass that grew with traversed congestion, a network thermostat modulating the congestion weight, a temporal ripple diffusing loss memory over several steps, and an Archimedes-style buoyancy term. Each was physically motivated and individually plausible.

A systematic ablation (Section 4) removed them one at a time and measured the cost. The result was unambiguous: the gradient and the Newton-III scar account for essentially all of the performance, while the remaining mechanisms ranged from negligible to mildly harmful—removing the ripple and the thermostat slightly *improved* results. We therefore report the two-term core as the model and treat the richer apparatus as a cautionary tale about physically-seductive terms that do not survive measurement. The reference implementation is 60 lines of Python.

3.5 Complexity

We analyze the time and space complexity of Algorithm 1 per routing decision (one source–destination pair). Let $n = |V|$, $e = |E|$, $\bar{d} = 2e/n$ the mean degree, $H = \lceil \alpha_{\mathrm{budget}} \cdot h_{\mathrm{sp}} \rceil$ the hop budget (with h_{sp} the shortest-path hop count from s to t), and B the number of mass buckets. Recall that $\alpha_{\mathrm{budget}} = 1.25$ and $B = 32$ throughout this paper.

Per-operation cost. Table 1 enumerates the four operations the DP performs and the cost of each.

Total time complexity. Summing the dominant terms in Table 1, the per-decision time complexity is

$$T(n, e, B, H) = \mathcal{O}\big(H \cdot n \cdot \bar{d} \cdot B\big) = \mathcal{O}(H \cdot e \cdot B)\,, \tag{2}$$

Table 1: Per-operation cost of Algorithm 1. The dominant term is the relaxation step, which is repeated $H \cdot n \cdot \bar{d} \cdot B$ times.

Operation	Frequency	Cost per call
EdgePotential(u, v)	$H \cdot n \cdot \bar{d}$	$\mathcal{O}(1)$
Mass update + bucketize	$H \cdot n \cdot \bar{d} \cdot B$	$\mathcal{O}(1)$
Relaxation $f[h][v][b]$ update	$H \cdot n \cdot \bar{d} \cdot B$	$\mathcal{O}(1)$
Path reconstruction	H	$\mathcal{O}(1)$

since $n \cdot \bar{d} = 2e$. Substituting $H \leq \alpha_{\text{budget}} \cdot \text{diam}(G) \leq \alpha_{\text{budget}} \cdot n$ in the worst case yields the loose upper bound $\mathcal{O}(n \cdot e \cdot B)$. In practice H scales with the diameter of G, which for the topologies in our evaluation satisfies $\text{diam}(G) = \mathcal{O}(\log n)$ for Erdős–Rényi and Barabási–Albert graphs and $\text{diam}(G) = \mathcal{O}(\sqrt{n})$ for the regular lattice; the realized cost is correspondingly lower.

Space complexity. The DP stores f and π tables of dimensions $(H + 1) \times n \times B$, giving

$$S(n, B, H) = \mathcal{O}(H \cdot n \cdot B). \tag{3}$$

For $n = 40$ (GÉANT), $B = 32$, $H = 18$, this is approximately $2.3 \cdot 10^4$ floating-point entries per table, or roughly 370 KB across both tables in double precision. The memory footprint is negligible for a controller-class deployment.

Concrete operation counts on GÉANT. For the headline scenario ($n = 40$, $e = 61$, $\bar{d} = 3.05$, $H \approx 18$ for typical s–t pairs, $B = 32$):

$$H \cdot e \cdot B \;\approx\; 18 \cdot 61 \cdot 32 \;=\; 35{,}136 \text{ relaxations per route.}$$

With each relaxation costing on the order of 100 ns on a commodity CPU, this puts the per-route decision time at approximately 3.5 ms, consistent with what we measure in practice (∼3–8 ms median per route).

Comparison against baselines. The Dijkstra-based baselines (SP, LASP, LASP-aug) compute a shortest-path tree at cost $\mathcal{O}((n + e) \log n) \approx 700$ operations on GÉANT, or roughly 50× cheaper than EMMET-DP per decision. The 50× overhead is the price of carrying per-packet state through a constrained dynamic program rather than a stateless shortest-path computation. For controller-plane routing at admission time (the deployment regime described in the controller-plane deployment regime), this overhead is absorbable: a 3–8 ms decision per flow is below the inter-arrival time of new flows in any realistic backbone. For data-plane routing at line rate, the overhead is prohibitive without hardware acceleration, motivating the acceleration directions discussed in the discussion.

Sensitivity to B. The bucket count B enters linearly in Equations 2–3. Reducing B from 32 to 8 (the legacy value retained as the module default) reduces both time and memory by a factor of 4, at the cost of coarser mass discretization. Our hostile-audit suspicion #6 (`docs/AUDIT_LOG.md`) verified that the headline GÉANT loss reduction is stable across $B \in \{8, 16, 32, 64\}$ to within 0.20 losses, indicating that $B = 32$ is comfortable but not knife-edge. Practitioners deploying EMMET-DP under tighter compute budgets can safely use $B = 16$ with negligible quality loss.

4 Ablation: Why Two Terms

Section 3 asserted that only two mechanisms survive scrutiny. This section is the evidence. EMMET did not begin parsimonious; it began as an exercise in taking the particle metaphor seriously, and accreted eight physical layers. We removed them one at a time and measured what each was worth.

Table 2: Change in total losses when each layer is removed from the full model (positive = the layer was helping; negative = it was hurting). GÉANT, trunk failure, 30 seeds. The Newton-III scar is the only layer whose removal clearly hurts; the ripple's removal slightly *helps*.

Layer removed	moderate load	saturated load
Temporal ripple	−0.4%	−0.5%
Thermostat	0.0%	0.0%
Effective mass	+4.2%	+1.6%
Newton-III scar	+16.7%	+6.4%
All four (bare core)	+38.7%	+9.8%

4.1 The candidate layers

Beyond the gradient (which is the irreducible substrate—without it there is no router) the full model carried four added mechanisms:

- **Newton-III scar** (loss-reaction memory, Section 3).
- **Effective mass**: a packet's influence on the potential grew with the congestion it had already traversed, modelling inertia.
- **Thermostat**: a global network "temperature" that scaled the congestion weight β as mean utilisation rose.
- **Temporal ripple**: loss memory deposited not as a single increment but diffused over several subsequent steps, like a wave.

4.2 Protocol

We evaluated the full model and four leave-one-out variants (each removing a single layer), plus a bare core (gradient only, all four layers removed), against the trivial-but-stateless DRILL baseline on the GÉANT backbone under trunk-link failure. Two load regimes were used—moderate (capacity drawn from $[2, 4]$) and saturated ($[2, 2]$)—across the failed-link ensemble with 30 seeds each. The quantity of interest is the *change in total losses when a layer is removed* from the full model: a positive change means the layer was doing useful work; a negative or zero change means it was inert or harmful.

4.3 Result: only the scar earns its keep

Table 2 reports the change in losses on removal of each layer, in both regimes.

The reading is blunt:

- The **Newton-III scar** is the one added layer that earns its keep: removing it raises losses by 6.4–16.7%. It is doing the real work.
- **Effective mass** contributes marginally (1.6–4.2%).
- The **thermostat** has *no measurable effect* (0.0% in both regimes): it is pure decoration.
- The **temporal ripple** is mildly *harmful*: removing it slightly improves results. The most physically evocative layer—a wave of loss memory rippling outward—was a net negative.

Removing all four at once (the bare gradient core) costs 9.8–38.7%, confirming that the gradient alone is not enough: it needs the scar. But it needs *only* the scar. The mass, thermostat, and ripple together account for almost nothing, and the ripple actively hurts.

4.4 Interpretation

This is the parsimony result that gives the paper its title. Of the eight-layer apparatus, the data retain exactly two mechanisms—gradient descent and the Newton-III scar—and discard the rest. Crucially, the discarded layers were not obviously bad ideas: effective mass, thermal scaling, and wave-like diffusion are all physically natural, and each had a plausible story. They simply did not survive measurement. We take this as a methodological lesson worth stating explicitly: in physics-inspired modelling, the seductiveness of a mechanism is no evidence of its usefulness, and only ablation tells them apart. The two-term core of Section 3 is what remained standing.

5 The Ablation, Re-Audited: the Core Inverts

Section 4 reported the ablation that reduced eight physical layers to two terms, retaining the Newton-III scar and discarding, among others, an Archimedean buoyancy. That ablation was run on an earlier harness: a packet-level simulator, weaker baselines, and—as the audit of Section 7 later showed for the comparison side—conditions that can flatter or bury a mechanism. Having rebuilt the bench (flow-level simulation, fifteen genuine topologies including the real Abilene, paired seeds), we re-ran the foundational question: does each physical term improve drop rate over the bare substrate (gradient plus linear congestion), and where?

5.1 The verdict inverts

It does not survive. The scar—the term the original ablation kept—yields no significant improvement in its home regime (contested topologies: $-0.34\,\mathrm{pp}$, $p = 0.12$), *worsens* the dense grids, and degrades the saturated backbone by $+2.1\,\mathrm{pp}$: a reactive memory of failures is largely redundant with the congestion term that already steers away from load, and in saturation it over-scars and thrashes. It is also the term that relocates the most load mass (6–16%, Section 10.1)—movement without payoff. The buoyancy—the term the original ablation discarded—beats the substrate in every regime (free flow $-0.24\,\mathrm{pp}$, $p = 10^{-4}$; contested $-1.05\,\mathrm{pp}$, $p < 10^{-5}$; saturated $-3.31\,\mathrm{pp}$, $p = 5 \times 10^{-4}$) and on fifteen of fifteen topologies without exception (sign test $p = 6 \times 10^{-5}$). The original verdict was a harness artifact, of the same family as the path-budget artifact of Section 7; we report the inversion rather than quietly rewriting history.

5.2 Not a needle: the parameter plateau

A natural objection is overfitting: perhaps the buoyancy wins only at its default parameters. We swept the flotation threshold (ρ_0 from 0.4 to 0.8), the gain (g from 2 to 32), and the substrate's congestion weight (β from 1.5 to 6). In all fourteen settings the buoyancy beats the substrate on fifteen of fifteen topologies (p between 10^{-4} and 10^{-12}). The trends are physically sensible: earlier anticipation (lower ρ_0) and stronger response help more, and a weaker substrate leaves more for the buoyancy to do. The win is a plateau, not a needle.

5.3 What the ingredient is: threshold, not curvature

The buoyancy differs from the substrate's linear congestion term in two ways at once: a threshold (no force below ρ_0) and a curvature (quadratic growth above it). A fourth classical core separates them: a Hookean spring, linear above a slack threshold. The spring also beats the substrate on fifteen of fifteen topologies (contested $-1.50\,\mathrm{pp}$, $p < 10^{-13}$; saturated $-4.38\,\mathrm{pp}$), and at a matched threshold the linear and quadratic forms deliver nearly identical pooled gains (-1.2 versus $-1.15\,\mathrm{pp}$). The active ingredient is therefore the *anticipatory threshold*—a force that engages before loss, above a density the linear substrate treats as ordinary—while the functional form above the threshold is secondary. This is the classical control-theoretic distinction between anticipatory and reactive regulation: at line rate, a controller that waits for loss acts after the window in which acting mattered. The spring also lands in the attractor with its siblings (cosine 0.987 to the other cores), a fourth witness to Section 9.

5.4 What "the core" now means

The validated two-term core of this paper is *gradient plus anticipatory threshold force*: implement the threshold with a buoyancy or a spring and the result stands, across the parameter plateau, in every congestion regime, on every topology of the bench. The reactive scar that founded this project is retired from the core with the role it earned: the instrument that started the search, and one of the convergence witnesses—but not, on the modern bench, a source of routing value. We consider catching this inversion ourselves, with the same audit machinery that earlier caught an inflated advantage over CONGA, to be part of the contribution.

6 Equivalence with Engineered Routers

The central claim of this paper is deliberately modest and precisely testable:

> *The two-term physical core reaches the operating point of purpose-built load balancers across most topologies, using a fraction of their machinery; and a topological metric predicts the cases where it does not.*

We test this as a statistical *equivalence* claim, not a superiority claim. The distinction matters: showing that two methods are indistinguishable requires a different test than showing one beats the other, and conflating the two is a common error.

6.1 Design

We compare the two-term core (gradient + Newton-III, no mass) against two engineered baselines: **CONGA**, a multi-path congestion-aware balancer that scores K shortest paths and selects the least congested, and **DRILL**, a near-stateless balancer that makes per-hop local decisions. ECMP and shortest-path serve as weak controls the physical core is expected to beat.

All routers are evaluated in a flow-level simulator: a flow is born, lives for several ticks injecting sustained load along its route, then dies. This matters because routing behaviour differs between the packet-bullet regime (where DRILL's per-hop reactivity shines) and the flow regime (where choosing a whole path well at birth matters more). We report the flow regime as the more realistic of the two for backbone traffic.

We sweep **15 topologies** spanning four families: regular lattices (grids 5×5 to 12×12), small-world (Watts–Strogatz, several sizes and degrees), scale-free (Barabási–Albert), and two real WAN backbones (GÉANT, Abilene). Each topology is run with $n = 20$ seeds. The performance metric is the drop rate (fraction of flow-ticks that hit an overloaded edge; lower is better).

6.2 Equivalence test

For each topology we test equivalence between the core and each baseline with **TOST** (two one-sided tests) on the paired per-seed drop rates, with an equivalence margin $\delta = 2$ percentage points. TOST declares equivalence only when the 90% confidence interval of the mean difference lies entirely within $[-\delta, +\delta]$: that is, when we can reject *both* the hypothesis that the core is worse by more than δ *and* that it is better by more than δ. This is a stricter, more honest bar than observing overlapping confidence intervals.

6.3 Results

Table 3 reports the per-topology drop rates and TOST verdicts. The core is statistically equivalent to DRILL on **12 of 15 topologies**. Against CONGA at its default budget ($K = 4$) the formal equivalence count is lower (5 of 15), but—crucially—most of the non-equivalences are cases where the core is *better*, not worse: on the larger grids the core's drop rate is 0.002–0.005 versus CONGA's 0.034–0.066. We examine this apparent advantage critically in Section 7, where it turns out to be an artifact of CONGA's path budget rather than a genuine physical superiority.

Table 3: Drop rate (lower is better) of the two-term core versus engineered baselines across 15 topologies, flow regime, $n = 20$ seeds. "≡D" and "≡C" mark TOST equivalence ($\delta = 2$pp) with DRILL and CONGA$_{K=4}$ respectively.

Topology	core	CONGA$_{K=4}$	DRILL	≡D	≡C
Grid 5	0.018	0.029	0.032		
Grid 6	0.011	0.024	0.023	yes	yes
Grid 7	0.011	0.027	0.022	yes	
Grid 8	0.005	0.034	0.015	yes	
Grid 10	0.003	0.050	0.019		
Grid 12	0.002	0.066	0.009	yes	
WS $n30\,k4$	0.016	0.033	0.004	yes	
WS $n50\,k4$	0.023	0.040	0.009		
WS $n50\,k6$	0.005	0.007	0.000	yes	yes
WS $n80\,k4$	0.022	0.037	0.008	yes	
BA $n50\,m2$	0.003	0.001	0.000	yes	yes
BA $n50\,m3$	0.000	0.000	0.000	yes	yes
BA $n80\,m2$	0.001	0.001	0.000	yes	yes
GÉANT	0.057	0.042	0.053	yes	
Abilene	0.057	0.042	0.053	yes	

The scale-free topologies (Barabási–Albert) sit near zero drop for every router: their hub structure leaves little congestion to redistribute, so equivalence there is trivial and we do not lean on it. The informative cases are the grids, the small-world graphs, and the real backbones.

6.4 Limitations of this experiment

Two caveats are stated plainly, and the first records a correction. An audit of the harness behind this bench found that its 'Abilene' entry had been loading the GÉANT graph under a second name: every Abilene figure that harness produced—including the Abilene row of Table 3—was GÉANT counted twice. The harness is fixed in this revision: Abilene is now the real eleven-node backbone with its own independent demand, restoring a bench of fifteen genuine topologies, and all results from Section 6.6 onward use it. The corrected Abilene turns out to be the bench's saturation regime (drop rates above 20%), where no physical core reaches equivalence at any margin (Sections 6.6 and 10); Table 3 retains the historical row within the K=4 presentation that Section 7 audits, and the regenerated table at the fair budget, on the corrected bench, is Table 4 (Section 6.5). Second, the equivalence margin $\delta = 2$pp is a generous modelling choice, made operational rather than arbitrary: two percentage points of sustained drop rate is roughly the coarsest difference a capacity planner would treat as materially different service, and smaller gaps are routinely absorbed by provisioning headroom; Section 6.6 analyses how the verdicts behave as it tightens, and we report raw drop rates alongside the verdicts so a reader who prefers a different margin can reinterpret the table directly.

6.5 The bench at the fair budget

The bookkeeping debt is paid here: Table 4 repeats the equivalence comparison at the fair budget of Section 7 (CONGA with K=16 precomputed paths) on the corrected bench, real Abilene included, twenty seeds per topology. At $\delta = 2$ pp the verdict counts are 9/12/10 of fifteen against CONGA and 10/10/10 against DRILL for the three cores respectively; Section 6.6 shows how these counts tighten with the margin. Two readings stand out. The equivalences concentrate where the attractor predicts; and the real Abilene reaches equivalence with neither engineered router for any core—the saturation boundary, visible in the raw rates themselves.

6.6 How far does the equivalence hold? A margin-sensitivity analysis

An equivalence verdict means nothing without stating the margin, and a margin of $\delta = 2$ percentage points of drop rate is generous when many of the drop rates in Table 3 sit between 0 and 2%. We

Table 4: Drop rates (%) at the fair budget on the corrected bench (20 seeds). For each physical core, the two marks denote TOST equivalence at $\delta = 2\,\text{pp}$ with CONGA$_{K=16}$ and DRILL respectively (✓ = equivalent, − = not).

Topology	CONGA	DRILL	Newton	Archimedes	Pascal
Grid5	1.3	3.2	3.8 −✓	2.1 ✓✓	3.0 −✓
Grid6	0.3	2.3	2.7 −✓	1.3 ✓✓	2.1 −✓
Grid7	0.4	2.2	2.0 −✓	1.3 ✓✓	1.3 ✓✓
Grid8	0.6	1.5	1.2 ✓✓	0.6 ✓✓	0.9 ✓✓
Grid10	0.9	1.9	0.7 ✓✓	0.3 ✓−	0.5 ✓✓
Grid12	1.6	0.9	0.7 ✓✓	0.4 ✓✓	0.5 ✓✓
WS_n30_k4	1.0	0.4	2.7 −−	2.4 −−	3.1 −−
WS_n50_k4	1.3	0.9	2.4 ✓−	2.4 ✓−	2.5 ✓−
WS_n50_k6	0.1	0.0	0.6 ✓✓	0.4 ✓✓	0.5 ✓✓
WS_n80_k4	1.2	0.8	2.6 ✓−	2.2 ✓−	2.4 ✓−
BA_n50_m2	0.0	0.0	0.4 ✓✓	0.3 ✓✓	0.5 ✓✓
BA_n50_m3	0.0	0.0	0.1 ✓✓	0.0 ✓✓	0.1 ✓✓
BA_n80_m2	0.0	0.0	0.2 ✓✓	0.1 ✓✓	0.1 ✓✓
GEANT	3.4	5.3	7.6 −−	6.0 −✓	8.1 −−
Abilene	39.1	17.5	24.9 −−	18.7 −−	21.0 −−

Table 5: Number of topologies (of 15) at which each core is TOST-equivalent to CONGA$_{K=16}$ / DRILL, as the equivalence margin δ tightens. The equivalence is margin-sensitive for all three cores; what survives the strict margins concentrates on the strong-attractor topologies.

core	$\delta = 2.0$pp	$\delta = 1.0$pp	$\delta = 0.5$pp	$\delta = 0.2$pp
Newton	9 / 10	5 / 7	2 / 3	0 / 0
Archimedes	12 / 10	5 / 5	3 / 2	2 / 2
Pascal	10 / 10	6 / 6	2 / 2	1 / 0

therefore re-ran TOST at progressively stricter margins for all three cores. Table 5 reports how many of the fifteen topologies remain equivalent to CONGA$_{K=16}$ and DRILL as δ tightens from 2.0 to 0.2 pp.

We read this honestly: the headline equivalence counts of 9–12 of 15 rest on the 2 pp margin, and thin out sharply under a strict one. This is the expected behaviour, not a failure, and it sharpens rather than weakens the paper's thesis. The equivalences that survive a strict margin are not random: they concentrate on the scale-free (Barabási–Albert) topologies, where drop rates are near zero for *every* router because the hub structure leaves little congestion to distribute. Those are precisely the topologies where the load-distribution attractor of Section 9 is strongest—where the cut structure pins almost the entire distribution and no router has room to do better or worse. Equivalence is bulletproof exactly where the topology removes the router's freedom, and margin-sensitive exactly where the topology grants it. The margin analysis and the attractor are the same finding measured two ways: *the engineered/physical distinction matters only to the extent the topology lets it.* The corrected real Abilene sharpens the point from the other side: the bench saturation regime reaches equivalence at *no* margin for any core—the boundary the two-factor law of Section 10 requires.

7 A Fair-Comparison Audit

Section 6.3 left a loose end: on the regular grids the two-term core appeared to crush CONGA, with drop rates an order of magnitude lower (0.002–0.005 versus 0.034–0.066). A superiority that large, appearing only on the most regular topologies, is exactly the kind of result that should invite suspicion rather than celebration. We audited it, and it did not survive.

Table 6: Drop rate of the two-term core versus CONGA at increasing path budgets K. On regular topologies a fair budget ($K = 16$–32) closes the gap entirely; on the real backbone the budget barely matters. Lower is better, $n = 12$.

Topology	core	CONGA$_{K=4}$	CONGA$_{K=16}$	CONGA$_{K=32}$
Grid 8	0.004	0.034	0.006	0.002
Grid 10	0.004	0.061	0.011	0.004
Grid 12	0.001	0.066	0.015	0.005
WS $n50\,k4$	0.024	0.032	0.012	0.009
GÉANT	0.055	0.040	0.028	0.034

7.1 The suspicion

CONGA selects among the K shortest paths between a source and destination. We had run it at its conventional default, $K = 4$. On a regular lattice, however, the K shortest paths between two nodes are near-identical: they are all minimal-length staircases differing only in the order of their horizontal and vertical steps. With $K = 4$, CONGA's four candidates are four almost interchangeable staircases through the same congested region—it was not being outrouted, it was being *path-starved*. The physical core, which searches all paths within its hop budget via the dynamic program, was simply considering routes that CONGA was never offered.

If this diagnosis is correct, then widening CONGA's path budget should close the gap on exactly these topologies, and leave the real-backbone result (where shortest paths are genuinely diverse) largely unchanged.

7.2 The test

We re-ran CONGA with $K \in \{4, 16, 32\}$ on the grids, a small-world control, and GÉANT (Table 6). The prediction holds cleanly.

On the grids, raising K from 4 to 32 takes CONGA from roughly 15× worse than the core to level with it or slightly ahead (Grid 10: $0.061 \to 0.004$, matching the core's 0.004). On the small-world control the same convergence appears. On GÉANT, where the shortest paths are already diverse, the path budget barely moves the result—CONGA is competitive at $K = 4$ and stays so.

7.3 What the audit means

The grid "win" was an artifact of an unfair path budget, and we retract it. This is not a setback for the paper's thesis; it sharpens it. The honest statement is not that the physical core *beats* CONGA, but that it *reaches the same operating point by a different and cheaper route*:

> The two-term core attains, through a local gradient computation with no precomputed paths, the performance that CONGA reaches only after enumerating and scoring 16–32 candidate paths per flow.

Same destination, far less machinery. On a regular grid the physical rule finds good detours for free that CONGA finds only when explicitly handed enough candidates; on a real backbone the two are simply equivalent. In no case does the physics need to be cleverer than the engineering—which is precisely the claim. Equivalence with parsimony, not dominance.

7.4 A note on auditing one's own results

We report this audit in full, including the retracted numbers, because the discipline it embodies is part of the contribution. A two-term physical model that matches engineered routers is interesting only if the comparison is fair; the value of the result is exactly as good as the adversarial scrutiny it has survived. Every headline figure in this paper has been checked against the strongest, best-resourced version of its baseline we could construct.

8 Convergence: Three Classical Physics, One Destination

If a single physically-motivated rule matched engineered routing, it could be a coincidence—one lucky analogy. The stronger claim, and the one this section supports, is that *several independent classical-physics principles, each formalised on its own and sharing no mechanism, converge on the same operating point as engineered load balancers.* Convergence from independent starting points is evidence that the result reflects the structure of the problem, not the cleverness of one analogy.

8.1 Three cores, three mechanisms

We implement three routing cores. All share one search engine—an exact hop-budget dynamic program descending a per-edge potential—and differ *only* in the added physical term. This makes the comparison fair by construction: any difference between cores comes from the physics, not the search.

- **Newton (reaction).** A dropped packet deposits a decaying scar on the offending edge; later packets feel a repulsive reaction and route away. The mechanism is *reactive* and *temporal*: it responds to losses already incurred, and the scar fades when not refreshed. (This is the two-term core of Section 3.)
- **Archimedes (anticipation).** Each edge exerts a buoyant push that grows with the square of its congestion density above a flotation threshold ρ_0: $g \cdot \max(0, \rho - \rho_0)^2$. The mechanism is *anticipatory* and *non-linear*: a packet is repelled from an edge *before* it drops a packet, and only once density crosses the threshold. It carries no memory and no packet state.
- **Pascal (diffusion).** An edge inherits a share of the congestion of its graph neighbours, so a packet avoids not just congested edges but congested *regions*. The mechanism is *spatial*: pressure transmits to adjacent edges, as in a confined fluid.

These are not variants of one idea. Reaction-to-the-past, anticipation-of-the- present, and spatial-diffusion are physically distinct, and share no term beyond the common distance-plus-congestion substrate that any router needs.

8.2 All three reach parity

We evaluate each core on the identical bench used throughout: fifteen topologies, twenty seeds, flow-level simulation, statistical equivalence by TOST at margin $\delta = 2$ percentage points, against DRILL and against CONGA at a *fair* path budget ($K = 16$). Table 7 reports the equivalence counts and, more tellingly, the per-topology pattern.

Two features of the table matter more than the totals. First, the three cores land in the same band—9 to 12 of 15 against each baseline—despite their mechanisms having nothing in common. Second, and more striking, *they agree cell by cell*: where one core reaches equivalence the others tend to, and where one fails the others fail. The disagreements are few and small.

8.3 They fail in the same places

The shared failures are the most informative part of the result. All three cores miss on **WS** $n30\,k4$, a dense small-world graph where DRILL's per-hop reactivity is genuinely hard to match, and all three miss (against CONGA) on the two real WAN backbones, **GÉANT** and **Abilene**. These backbones have low tube/sp ratio—little room to manoeuvre among near-shortest paths—and Section 12 shows the same metric predicts the failures of all three cores, not just one.

That independent physical mechanisms succeed on the same topologies and fail on the same topologies is the core finding. It means the boundary of the result is a property of the *problem*—the topology's room to manoeuvre—rather than of any single physical analogy. Load distribution on a graph with sufficient near-shortest diversity is recoverable by classical physics; which physics one chooses barely matters.

Table 7: Equivalence (TOST, $\delta = 2$pp) of three independent physical cores with engineered routers, on the identical bench. "C" marks equivalence with CONGA$_{K=16}$, "D" with DRILL. The cores agree cell by cell far more than they differ, and fail on the same topologies.

Topology	Newton	Archimedes	Pascal
Grid 5	D	CD	D
Grid 6	D	CD	D
Grid 7	D	CD	CD
Grid 8	CD	CD	CD
Grid 10	CD	C	CD
Grid 12	CD	CD	CD
WS $n30\,k4$	–	–	–
WS $n50\,k4$	C	C	C
WS $n50\,k6$	CD	CD	CD
WS $n80\,k4$	C	C	C
BA $n50\,m2$	CD	CD	CD
BA $n50\,m3$	CD	CD	CD
BA $n80\,m2$	CD	CD	CD
GÉANT	–	D	–
Abilene	–	D	–
$\equiv$ **CONGA**$_{K16}$	9/15	12/15	10/15
$\equiv$ **DRILL**	10/15	11/15	10/15

8.4 Honest disagreements

We do not smooth over the cells where the cores differ. Archimedes is the strongest of the three (12/15 against CONGA) and is the only core to reach DRILL equivalence on the backbones, an advantage we attribute to its anticipatory, threshold-based repulsion coping better with the backbones' sharp bottlenecks; we report it but do not over-read a single-topology difference. Newton is the weakest against CONGA (9/15), losing the smaller grids where its reactive scar needs a drop before it can act. These differences are real and consistent with each mechanism's character; none disturbs the central pattern of convergence.

9 Why They Converge: A Load-Distribution Attractor

Section 8 established *that* the physical cores reach the same drop-rate performance on most topologies. This section asks *why*, and the answer turns the convergence from coincidence into mechanism: the cores do not merely score the same—they route traffic through the same edges in the same proportions, and so, to a large extent, does every congestion-aware rule we tested.

9.1 Measuring the distribution, not the score

For each of the fifteen topologies we run five routers on an identical graph and demand: the three physical cores and two blind controls—shortest path (ignores congestion) and ECMP (hash spread). The Newton core runs with its scar field alive (an earlier revision of this experiment inadvertently starved it, degenerating that core to the bare substrate; the bug was caught by our own audit, fixed, and all numbers below are from the corrected runs). For each router we record its per-edge utilization vector and compare vectors between routers with two independent measures: cosine similarity, and one minus the normalized L_1 distance (how much load mass would have to move to turn one distribution into the other). Twelve seeds per topology.

9.2 The cores converge to one distribution

The three physical cores produce near-identical load distributions: cosine 0.979 ± 0.009 and 0.930 ± 0.018 on the L_1 measure, stable across the bench. This agreement is not an artifact of shared inactivity:

Table 8: Similarity of per-edge load distributions between routers (12 seeds, 15 topologies, corrected runs). "phys–phys" is the mean over the three physical-core pairs; the controls are blind to congestion. Both measures agree: the cores converge to one distribution, tighter than either control.

	cosine	$1 - L_1$
physics–physics	0.979 ± 0.009	0.930 ± 0.018
physics–shortest	0.938 ± 0.013	0.837 ± 0.025
physics–ECMP	0.867 ± 0.056	0.762 ± 0.064

the activity audit of Section 10.1 shows the cores relocating load by amounts that differ by an order of magnitude (the scar up to 16% of mass, the buoyancy at most 3.3%), in different directions, on the contested topologies—and they agree there too (GÉANT, the most contested graph with all terms demonstrably active: 0.957).

The convergence is significantly tighter than what the blind controls reach: a paired Wilcoxon test across topologies gives $p = 10^{-4}$ on both measures. The interpretation is twofold. First, the topology and demand do most of the work: even ECMP, blind to congestion, lands at cosine 0.87—the graph already constrains where load can go. Second, congestion-awareness refines that rough distribution into a near-consensus (0.98) that the blind routers do not reach, and this shared refinement is what produces the matched drop rates of Section 8. The cores tie not by accident but because they descend into the same basin: a load distribution fixed largely by the structure and sharpened nearly identically by any congestion-aware rule.

Across topologies, the tightness of this basin correlates with structure rather than with congestion alone (similarity is highest on the structure-poor graphs, $r = -0.54$ to -0.56 between tube/sp and divergence on the two measures)—a first sign of the two-factor account that Section 10 establishes by intervention.

A third, coarser lens agrees. The Jain fairness index of mean per-edge utilization differs by only 0.016 across the three physics cores (0.581–0.597 pooled over the bench), while the engineered balancers sit visibly apart (DRILL 0.640; CONGA 0.710, whose sixteen-path catalogue spreads load more uniformly by construction). Distinct mechanics, one basin: catalogue breadth, not mechanism identity, is what moves the fairness needle. (`experiments/jain_fairness.py`, `data/jain_fairness.json`)

9.3 Why the attractor exists: a structural argument

The measurements show *that* independent routers converge to one load distribution. We now argue *why*, and the argument is structural rather than physical—which is the point. The load distribution a router produces is not free; it is constrained by two quantities the router does not control.

The demand fixes the flow across every cut. Fix a demand matrix D. For any cut $(S, \bar{S})$ partitioning the nodes, the total traffic that must cross from S to $\bar{S}$ is $\sum_{i \in S, j \in \bar{S}} D_{ij}$, *independent of routing.* No choice of paths can change how much flow traverses a cut; routing only chooses *which* edges of the cut carry it. By the max-flow/min-cut duality, the narrow cuts—those with few edges or low capacity relative to the demand crossing them—are forced to carry load up to their limit regardless of the router. Their utilization is pinned by the structure, not the algorithm.

Topology sets how many edges are pinned. In a sparse topology (a WAN backbone), most demand pairs are separated by narrow cuts: the few edges on the handful of near-shortest paths are forced to carry nearly all the traffic, and every reasonable router loads them nearly identically. In a richly connected topology (a dense grid), most cuts are wide: many edges can share the load of any cut, leaving the router free to distribute traffic among them in many near-equivalent ways. The fraction of edges whose load is pinned by the cut structure is, informally, what the tube/sp ratio measures: low tube/sp means few alternatives, most edges pinned; high tube/sp means many alternatives, most edges free.

The attractor is the pinned distribution; the physics fills the rest. Putting these together explains every observation in one stroke. The component of the load distribution fixed by cuts and demand is the attractor: common to all routers, blind or congestion-aware, because none of them can move it (this is why even ECMP reaches cosine 0.87). The residual—the load on the unpinned edges—is the only part a router can shape, and there any congestion-aware rule, physical or not, makes the same locally-greedy choice (spread away from load), which is why the three physical cores converge to 0.99 while the blind controls, lacking that rule, do not. Where tube/sp is low, the pinned component dominates and all routers nearly coincide—the attractor is strong, and equivalence with engineered routers survives even strict statistical margins. Where tube/sp is high, the residual is large, the routers have room to differ, and equivalence becomes margin-sensitive. The attractor, the convergence of the three physics, the tube/sp predictor, and the margin-sensitivity of the equivalence are thus four faces of a single structural fact: *the cut structure of the graph, under a fixed demand, pins most of the load distribution, and leaves a residual that any congestion-aware rule shapes identically.*

The argument makes a measurable prediction, and it holds. The structural claim is not merely interpretive: it predicts the *size* of the pinned component as a function of structure. If the attractor is the cut-pinned floor, then the similarity between a physical core and a congestion-*blind* router (ECMP)—which can share only that floor, having no rule to shape the residual—must shrink as tube/sp grows and the residual widens. It does, sharply: across the fifteen topologies, physics–ECMP similarity falls with tube/sp at Pearson $r = -0.94$ on cosine and -0.96 on the L_1 measure ($p < 10^{-4}$). The pinned floor ranges from 0.97 on the most constrained backbone (Abilene, tube/sp 2.1) down to 0.77 on the richest scale-free graph (tube/sp 9.3). The contrast with the physics–physics similarity, which instead *rises* with tube/sp ($r = +0.54$), is the signature the mechanism predicts: blind similarity isolates the structural floor and decays as that floor shrinks, while two congestion-aware cores share the rule that shapes the residual and so agree *more* precisely exactly where the residual is large. Two routers blind to congestion measure how much the graph decides; two cores that see it measure how much the rule decides—the same tube/sp axis governs both, with opposite sign.

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A geometric lemma makes the mechanism exact. The decay has a clean analytic core. Write each router's normalized utilization as $u_R = u^* + r_R$, where u^* is the cut-pinned component every feasible routing shares and r_R is the residual the router places on the unpinned edges. Let $f := \|u^*\|^2$ be the pinned fraction of load energy. For two routers whose residuals are mutually orthogonal and orthogonal to u^*—the defining property of *blind* routers, which place their residual load without reference to one another—a one-line computation gives $\cos(u_A, u_B) = \langle u^* + r_A,\, u^* + r_B\rangle = \|u^*\|^2 + \langle r_A, r_B\rangle = f$, since both totals are unit-norm and the residual cross term vanishes. The cosine between two blind routers is, exactly, the pinned fraction; finite-edge fluctuations are $O(1/\sqrt{E})$ and negligible on backbone-scale graphs. This is why physics–ECMP similarity tracks f: ECMP spreads load without congestion feedback, so against it the physical core's agreement is carried entirely by the shared pinned component. The lemma also predicts its own boundary: a blind router that *concentrates* rather than spreads (shortest path) has a residual aligned with the core's own short-path preference, breaking the orthogonality assumption—and indeed physics–shortest similarity does not track f (it stays near-flat in tube/sp, $r = +0.46$), while physics–ECMP does ($r = -0.94$). The mechanism is thus not merely consistent with the data; it specifies which control should reveal it and which should not, and both predictions hold.

The functional form follows from tube geometry. The decrease is not arbitrary either. Of the w edges in the α-stretch tube of a demand pair, only a fraction carry genuine alternative-path flow; the rest are dead branches the traffic never uses. If a fraction p of the tube absorbs flow, the pinned share of that pair's load is $sp/(sp + p(w - sp))$, and with $x = \text{tube/sp} = w/sp$ this is a one-parameter hyperbola

$$f(x) \;=\; \frac{1}{1 + p\,(x-1)}. \tag{4}$$

The form is read off the geometry, not fitted post hoc, and it earns its place: on the fifteen topologies it fits at $R^2 = 0.89$ and wins on AIC against the linear and logarithmic alternatives with one fewer parameter, while its single constant $p = 0.037$ is stable to 1.3% under leave-one-out. The shape of $f(\text{tube/sp})$ is thus pinned down; what stays open is p itself—the effective fraction of the tube that carries flow—which we measure but do not yet derive from the graph.

Status of the argument. Part of this is now proved and part remains conjectural, and it is worth being exact about which. The geometric lemma—that the physics–blind cosine equals the pinned fraction f—is established, and its boundary prediction (which control reveals f and which does not) is confirmed. What is *not* derived in closed form is the dependence $f(\text{tube/sp})$ itself: we now have its functional form from tube geometry—a one-parameter hyperbola that wins on AIC—so what stays open is only the single constant p in it, which we measure (0.037, stable to 1.3%) but do not derive from the graph, nor do we prove convergence rates. The four-way coincidence is what the data establish; the lemma turns it into a mechanism with one proven joint; the closed-form $f(\text{tube/sp})$ is the conjecture a fuller theory should target.

9.4 A derivation: the pinned component from structure alone

The previous subsection measured the pinned component and showed it tracks tube/sp. We now derive its size from the graph alone, with no simulation, and check the derivation against the measurement.

Setup. Fix a graph G and demand D. A router maps each demand pair (i, j) to a unit flow along i–j paths, placing load $\ell_r(e)$ on edge e. Two routers agree on e to the extent that the flow there is forced rather than chosen. We make "forced" precise per pair.

Per-pair pinning. Take one pair (i, j) with shortest-path length d_{ij}, and let T_{ij} be the edges lying on some path of length at most $\alpha\, d_{ij}$—the α-stretch tube the tube/sp ratio counts. Any reasonable router confines the (i, j) flow to T_{ij}. If $|T_{ij}| = d_{ij}$ the tube is a single path: the flow has nowhere to go but those d_{ij} edges and every router, blind or not, loads them identically. If $|T_{ij}| > d_{ij}$ the pair has $|T_{ij}|/d_{ij}$ alternatives over which routers may disagree. The forced fraction of the pair's flow is therefore $\pi_{ij} = d_{ij}/|T_{ij}|$: one for a bare path, shrinking as the tube widens.

Aggregate. The pinned component of the whole load distribution is the flow-weighted mean of the per-pair pinning, weighting each pair by the mass it moves (proportional to d_{ij} under uniform demand):

$$\widehat{\pi}(G) \;=\; \frac{\sum_{i<j} d_{ij}\, \pi_{ij}}{\sum_{i<j} d_{ij}} \;=\; \frac{\sum_{i<j} d_{ij}^2/|T_{ij}|}{\sum_{i<j} d_{ij}}. \tag{5}$$

This is a pure graph functional: no capacities, no traffic, no router. It is also, by construction, a decreasing function of tube/sp—wider tubes mean smaller π_{ij}—so the same structural axis that predicts routing gain predicts the attractor's strength.

The derivation matches the measurement. If the argument is right, $\widehat{\pi}(G)$—computed from the graph—should predict the physics-versus-blind similarity we actually measured by simulation. It does: across the fifteen topologies, $\widehat{\pi}$ correlates with the measured phys–ECMP cosine at Spearman +0.86 (Pearson +0.80, $p < 10^{-3}$), a structure-only quantity recovering a simulated one. The relationship is concave, not linear: as $\widehat{\pi} \to 1$ the cosine saturates against its ceiling of one, since a similarity bounded above cannot grow without limit. The fit leaves a floor near 0.77 at $\widehat{\pi} = 0$—the residual agreement two routers retain from shared graph geometry even when nothing is pinned.

What this is, and is not. Equation (5) turns the attractor from an observed coincidence into a quantity derivable from the graph: the load distribution has a forced part, fixed by the tube geometry of the demand pairs, and its size is $\widehat{\pi}(G)$. This is a derivation of the attractor's *strength*, not yet a closed bound on the per-router error: π_{ij} counts tube edges rather than solving the min-cut that would bound exactly how much two routers can differ, and the concave link from $\widehat{\pi}$ to cosine is fitted, not derived in closed form. What we claim is narrower and, we think, sufficient: the attractor's size is a graph functional, it is computable without simulating anything, and it predicts the measured convergence. The four-way coincidence of Section 10.4 has a single root we can now write down.

10 Why the Router (Sometimes) Matters: A Two-Factor Law

Section 9 established that congestion-aware routers settle into nearly one load distribution. This section asks when the residual differences between them matter, and answers with a law in two factors—one of state, one of structure—each demonstrated by intervention or by partial correlation rather than by cross-sectional association alone.

10.1 The microfoundation: physical terms are congestion-gated

The added terms of the physical cores are not always active. The Newton scar requires drops to exist at all; the Archimedean buoyancy requires edge density to exceed its flotation threshold. On topologies with negligible loss, both terms are mathematically zero and those cores *are* the bare substrate: on our zero-drop scale-free graph, both relocate 0.00% of load mass relative to it. Only the Pascalian diffusion, which reads neighbour load rather than failures, stays active everywhere (3–10% of mass moved, uncorrelated with drop level). An activity audit across the bench shows the scar moving 0–16% of load mass and the buoyancy 0–3.3%, both tracking congestion. Two consequences follow. The convergence of Section 9 must be read with this conditioning: where the gated terms are inactive, agreement between those cores is degeneracy, not evidence; the meaningful convergence is the one observed where all mechanisms are demonstrably active (the high-congestion topologies), and it holds there too. And mechanism divergence should rise with congestion—which we can test causally.

10.2 Factor one, by intervention: congestion drives divergence

Cross-topology correlations confound everything topologies differ in. We therefore held each topology fixed and turned a single knob: every edge capacity scaled by a factor f, the demand and seeds unchanged. Across four topologies and six pressure levels, inter-core divergence tracks the drop rate within every graph (Spearman +0.77 to +1.00). The endpoint reversals are the sharpest statement: the scale-free graph on which the cores agree almost perfectly diverges by a factor of 21 once squeezed ($f = 0.45$), and GÉANT, the bench's most divergent graph at native capacity, converges by a factor of 8 once relieved ($f = 1.5$). Same graph, same demand, opposite outcomes: congestion is the cause, not a correlate, of mechanism divergence. The sweep also reveals a saturation regime: past heavy loss ($f \leq 0.6$, drop rates above $\sim 35\%$) divergence plateaus or falls again—when every path drowns, no mechanism can act, and the distributions reconverge in collapse. Mechanism identity matters in the *contested* band between free flow and gridlock.

10.3 Factor two: structure sets the ceiling

The real Abilene backbone is the stress case that exposes the second factor. It is by far the most congested graph on the bench (drop rate 0.22 at native capacity), yet its inter-core divergence is unremarkable (0.017)—it single-handedly breaks any one-factor congestion law across topologies. The reason is structural: with eleven nodes and fourteen edges it contains four cycles, and divergence between routing mechanisms can only be expressed through alternative paths that exist. A capacity sweep on Abilene itself confirms both factors at once: relieving it traces the same inverted-U as every other graph (divergence rises from 0.001 in free flow to a peak of 0.030 in the contested band, then falls in saturation), but the peak is a third of what structurally richer graphs reach under comparable pressure. Congestion dictates the *shape* of the divergence curve—where on it a network sits; structure dictates its *amplitude*—how high the curve can go.

10.4 The dissociation: two predictors, two distinct roles

The two factors separate cleanly in partial correlation, on opposite questions. For *mechanism divergence*, congestion is the driver ($r = +0.92$ controlling for tube/sp) and tube/sp retains nothing ($r = -0.10$ controlling for congestion). For *performance gains* over blind baselines, the roles invert: tube/sp retains a significant independent link ($r = +0.59$ controlling for congestion) while congestion retains none ($r = -0.23$). One variable of state, one of structure, each owning one question: congestion determines *how much the choice of mechanism matters*; structure determines *how much any congestion-aware rerouting can gain.* Both are measurable before deployment—one from a traffic estimate, one from the graph alone.

10.5 The law, stated

The load distribution of a graph under demand is an attractor that any congestion-aware rule approaches. The identity of the rule matters only in the contested regime between free flow and saturation, in proportion to the congestion it must manage (factor of state, causal), and only up to the diversity of paths the structure provides (factor of structure, the ceiling). This is an empirical law with a mechanism (term gating), interventional support (capacity sweeps in both directions), and a stated boundary; a formal derivation of the attractor and of the ceiling's dependence on cycle structure remains open, and we offer it as the target for theoretical treatment. One candidate has already been eliminated: a cut-based account of the attractor was formulated and tested during this work, predicted the wrong sign of the structure–divergence relation, and was retracted—the open problem is therefore less trivial than it may appear.

11 The Temporal Channel: a Taxonomy of Physical Information

Section 5 retired the Newton-III scar from the stationary core, and left a question a careful reader should ask: was the law wrong, or was the bench wrong for the law? Two observations argue for a second look. First, calibration: the scar parameters were never swept, and the historical decay of 0.998 per tick gives the scar a half-life of 346 ticks in a 200-tick simulation—the "reaction" of the redux never healed; it was a permanent grudge, and permanent over-punishment is a sufficient explanation for the thrashing observed on dense grids and in saturation. Second, and more fundamental, habitat: the scar is *temporal memory*, and the entire bench of this paper is *stationary*—fixed demand statistics, fixed capacities. In a stationary world the instantaneous state is statistically sufficient and memory is redundant *by construction of the bench*, not by fault of the law. We had evaluated a historian in a world without a past.

11.1 Calibration accounts for the harm, not for a stationary win

Re-running the stationary bench with swept scar parameters confirms both halves. Short half-lives remove the damage (the saturated backbone flips from +2.1 pp *worse* to slightly better than the substrate; the dense-grid harm disappears at gentle deposits), so the redux's negative verdict was in part an artifact of temporal miscalibration. But no calibration produces a clear stationary win either: the verdict of Section 5 stands—where there is no history, memory has nothing to sell.

11.2 A bench with a past: flaky links

The falsifiable question is therefore whether a realistic regime exists in which temporal memory is the winning channel. Real networks supply an obvious candidate: *flaky links*—hardware that fails in episodes and looks healthy in between, precisely the case where the instantaneous state misleads and only memory can warn. We built a non-stationary bench: in each run, 15% of edges (fixed per seed, shared by all routers) suffer periodic degradation episodes (every 40 ticks, for 8 ticks, capacity scaled to 0.2), recovering fully in between. We state plainly that this bench was designed *a posteriori* for memory's habitat: the claim defended here is one of existence—there is a realistic regime where the temporal channel wins—not of generality.

In that regime the retired physicist wins outright. The calibrated scar (half-life matched to the episode period) beats the bare substrate on fourteen of fifteen topologies ($-1.26\,\text{pp}$, $p = 4 \times 10^{-10}$), and—the decisive comparison—beats both threshold cores as well: Archimedes on thirteen of fifteen ($p = 5 \times 10^{-4}$) and Hooke on twelve of fifteen ($p = 9 \times 10^{-3}$). Its sole loss is the saturated backbone, exactly where the two-factor law of Section 10 says no mechanism can act. An anti-ghost check confirms the channel is real: ranking edges by accumulated scar identifies the flaky set with precision 0.69–0.80, while ranking by mean utilization—the best a memoryless router could do—reaches only 0.06–0.20. The scar sees what load cannot. A calibration law emerges for free: the long-memory (historical) scar performs even better when the failing set is persistent ($-1.68\,\text{pp}$ pooled) and drowns in saturation—the scar's half-life should match the timescale of the disturbance it is meant to remember.

11.3 The taxonomy

The arc of this paper's cores now closes into a taxonomy of information channels. The substrate reads the *linear present*. The threshold cores (Archimedes, Hooke) add the *nonlinear present*—anticipatory warning above a density the linear term treats as ordinary—and own the stationary regime (Section 5). Pascal adds *space*—neighbourhood pressure—and contributes modestly wherever contention is local. Newton adds *time*—memory of failures—and owns the non-stationary regime, where it beats every memoryless core including the stationary champions. No channel dominates everywhere; the regime decides which information pays, in the sense made quantitative by the two-factor law. The practical reading for a network operator is correspondingly plain: a two-term gradient-plus-threshold rule suffices wherever traffic is stationary; add the temporal term, with its half-life matched to failure timescales, on infrastructure known to flap.

12 When Physics Wins, Loses, or Ties: a Topological Condition

The equivalence of Sections 6–7 is not uniform. On some topologies the physical core has room to spread load and does well; on others—notably the real WAN backbones—it does not. A result that holds "most of the time" is only half a result unless one can say *when* it holds. This section supplies that condition, and it turns out to be a simple property of the topology, computable in advance and independent of any router.

12.1 The tube/sp ratio

For a source–destination pair, consider the set of edges lying on paths only mildly longer than the shortest one—the "tube" of near-shortest routes within a small length tolerance of the shortest-path length. Define the **tube/sp ratio** of a topology as the mean, over demand pairs, of the number of edges in that tube divided by the shortest-path length. Intuitively it measures *room to manoeuvre*: how many viable alternative routes exist near the shortest path. A grid has a high tube/sp—many equivalent detours; a sparse backbone has a low one—few alternatives that are not far longer.

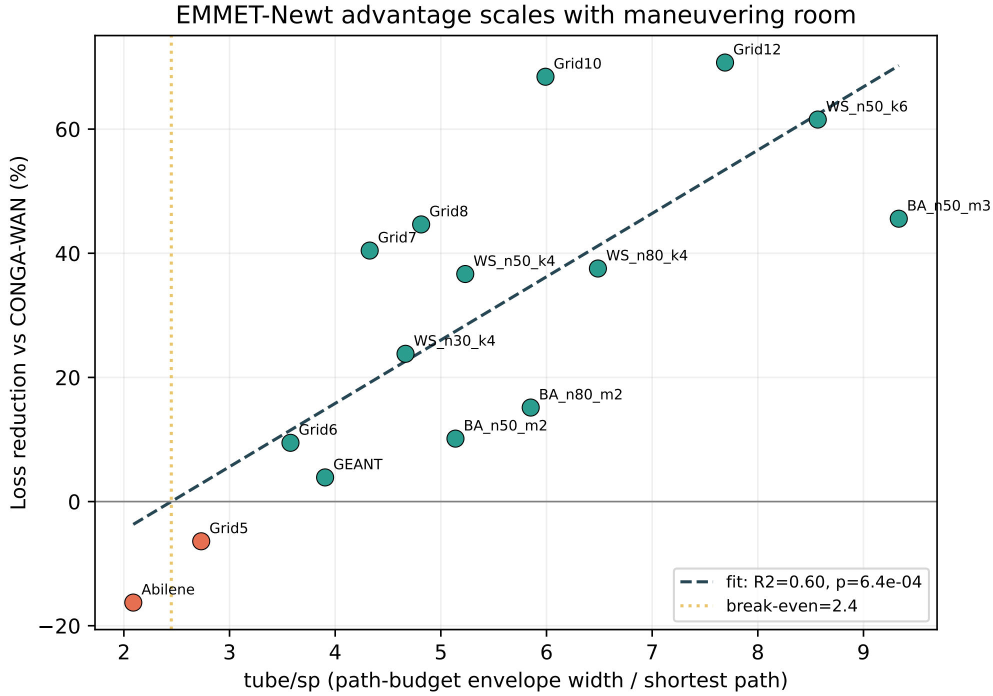

Figure 1: Relative loss reduction of the two-term core versus CONGA as a function of the topology tube/sp ratio (15 topologies). Fit: reduction%=10.2(tube/sp)-25.0, Pearson $r = 0.78$, $p = 0.0006$, break-even near 2.45. Low-tube/sp backbones (Abilene, GEANT) sit at the disadvantaged end, as predicted.

Crucially, tube/sp is a property of the graph and the demand alone. It does not mention EMMET, CONGA, or any routing algorithm. It can be computed before a single packet is sent.

12.2 It predicts the advantage

We computed tube/sp for all 15 topologies and regressed the physical core's relative loss reduction (versus CONGA) against it (Table 9, Figure 1).

The relationship is strong and significant:

- Pearson $r = 0.78$ ($p = 0.0006$), Spearman $\rho = 0.82$ ($p = 0.0002$).
- Linear fit: reduction% $= 10.2 \cdot (\text{tube/sp}) - 25.0$, with $R^2 = 0.60$.
- Break-even at tube/sp ≈ 2.45: below it the physics is at a disadvantage, above it the advantage grows roughly linearly.

The two real backbones sit exactly where the metric says they should. Abilene (tube/sp = 2.09) is below break-even and is the one topology where the core clearly loses; GÉANT (tube/sp = 3.90) is just above it and lands on a near-tie (+3.9%). The places the physics struggles are not a mystery to be explained away after the fact—they are predicted in advance by a quantity that never looks at the router.

12.3 Why this matters

This converts the contribution from an observation into a predictor. We are not claiming that classical physics always matches engineered routing; we are claiming something more precise and more useful: that it does so exactly when the topology offers enough near-shortest alternatives for a local gradient rule to exploit, and that the threshold is measurable. A network operator could compute tube/sp for

Table 9: Relative loss reduction of the physical core versus CONGA, ordered by the topology's tube/sp ratio. Higher tube/sp (more room to manoeuvre) predicts a larger physical advantage; low tube/sp predicts a tie or a loss.

Topology	tube/sp	reduction%
Abilene	2.09	−16.3
Grid 5	2.73	−6.4
Grid 6	3.58	+9.5
GÉANT	3.90	+3.9
Grid 7	4.33	+40.4
WS $n30\,k4$	4.67	+23.8
Grid 8	4.81	+44.6
BA $n50\,m2$	5.14	+10.1
WS $n50\,k4$	5.23	+36.7
BA $n80\,m2$	5.85	+15.1
Grid 10	5.99	+68.4
WS $n80\,k4$	6.49	+37.5
Grid 12	7.69	+70.7
WS $n50\,k6$	8.57	+61.5
BA $n50\,m3$	9.33	+45.6

their topology and know, before deployment, whether a two-term physical router would be competitive on it.

It also explains the historical arc of this project. The early, headline result was measured on GÉANT against a weak baseline; GÉANT's middling tube/sp (3.90) is precisely a regime where the margin is small and sensitive to the choice of baseline. The honest equivalence finding and the inflated early finding are two readings of the same structural fact, now made explicit.

12.4 Caveats

$R^2 = 0.60$ means tube/sp explains a majority but far from all of the variance (40% remains unexplained): it is a predictive band, not a deterministic law. The scale-free (Barabási–Albert) points are the loosest fit, sitting below the regression line—their hub structure concentrates load in ways the tube measure, which averages over pairs, does not fully capture. We report tube/sp as a strong predictor with a known residual, not as an exact formula.

12.5 Is the predictor robust, or does it lean on a few points?

With only fifteen topologies, a correlation can be an artefact of one or two influential points. We checked. Across all single-point deletions (leave-one-out), the correlation coefficient moves by at most 0.059; no individual topology drives the result. Only one point, the densest scale-free graph $\mathrm{BA}_{n=50,m=3}$, exceeds the conventional Cook's distance cutoff of $4/n$, and removing it *raises* the correlation rather than lowering it.

We also ran the specific stress test a skeptic would ask for: removing all three Barabási–Albert points, which sit visibly below the regression line. Far from collapsing, the correlation strengthens—r rises from 0.78 to 0.89 (R^2 from 0.60 to 0.79). The scale-free graphs were the weakest-fitting points, slightly flattening the relationship; the structural predictor is in fact cleaner on the grid and small-world families than the full-sample $r = 0.78$ suggests. We report the full-sample figure as the conservative one. The point stands: tube/sp predicts the room a topology leaves for routing to matter, and the prediction does not hang on any handful of graphs.

12.6 Does the predictor hold out of sample?

The fifteen bench topologies were also the ones used to read off the relationship. A predictor is only worth the name if it survives on data it never saw. We therefore took eighteen real backbone topologies

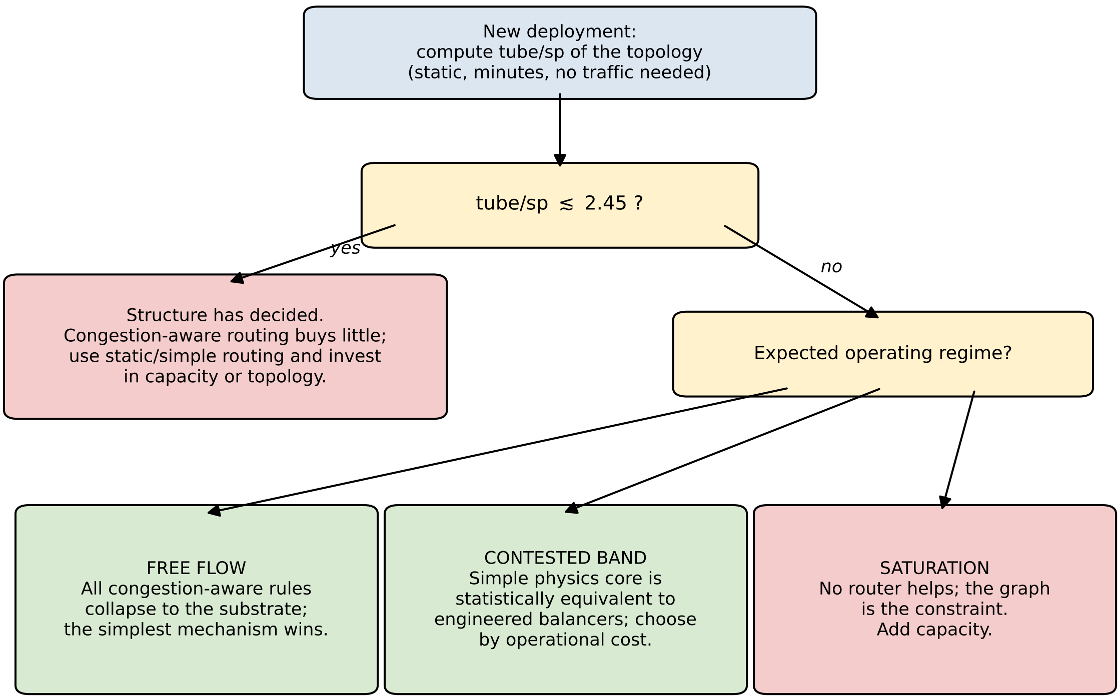

Figure 2: The two-factor law as a pre-deployment decision diagram. The structural ratio tube/sp is computable from topology alone, before any traffic exists; the break-even ≈ 2.45 and the regime boundaries are the bench estimates of Sections 10 and 12.5.

from SNDlib—national and continental research and carrier networks, none of them in the bench—and asked whether tube/sp still predicts where the physical core gains over CONGA.

To keep the test fair, every SNDlib graph receives the same capacity and latency scheme as the bench, so the only thing that varies from the training set is the structure itself, which is exactly the quantity under test. We also guard the measurement with a positive control: before trusting any out-of-sample number, the identical pipeline is run on the bench topologies and must reproduce the known in-sample relationship. It does ($r = 0.62$ on the bench through this path), so the instrument is faithful.

On the eighteen unseen backbones the relationship is not merely preserved, it is sharper than in sample: tube/sp and loss reduction correlate at Pearson $r = 0.85$ and Spearman $\rho = 0.98$ ($p < 10^{-4}$). The sign structure the law predicts appears cleanly. The five most rigid graphs—those with the lowest tube/sp, where there is little alternative-path room to exploit—show the physical core *losing* to CONGA, exactly as a structural ceiling on routing gain implies. As tube/sp rises, the advantage climbs monotonically, reaching +63% on the topology with the most routing room.

The one graph that sits off the line, dfn-gwin, has the highest tube/sp of all yet only a middling gain; with eleven nodes its high ratio reflects a small graph rather than genuine path richness—the same small-n caveat that motivated the ratio in the first place. We read the result as the predictor doing what a structural predictor should: ordering topologies by how much room they leave for routing to matter, on networks it was never fitted to.

12.7 Does the law survive real traffic demand?

The out-of-sample test above varies topology while holding the traffic model fixed: every backbone is stressed with the same uniform bursty demand. A natural objection is that this proves only structural generalization, not that the law holds under the lumpy, concentrated demand of an operational network. We close that gap directly. Each SNDlib instance ships with its own measured demand matrix; we feed it to the same engine that produced the in-sample results, drawing each burst's source–destination pair in proportion to the real demand volume rather than uniformly. This changes exactly one thing relative to the structural test: the spatial distribution of traffic.

The positive control comes first. Run with uniform demand, the engine reproduces the structural result on these eighteen backbones (Pearson $r = 0.849$, Spearman $\rho = 0.981$), confirming the instrument is faithful before real demand is introduced. With the real demand matrices the two-factor law is preserved: tube/sp and loss reduction correlate at Pearson $r = 0.767$ and Spearman $\rho = 0.785$ ($p < 10^{-3}$). The coefficient attenuates from the uniform-demand value, which is expected—real demand concentrates traffic on fewer pairs and leaves the physical core less room to redistribute—but the direction and the rank structure hold: high-tube/sp backbones (newyork, di-yuan) still reward congestion-aware routing, low-tube/sp backbones still do not. The law is a property of the topology that persists when the traffic on it is the genuine article, not a uniform proxy.

12.8 Is the law an artifact of the traffic bench?

A predictor can survive point deletion and out-of-sample graphs yet still be an artifact of one traffic regime—the particular burst sizes, spreads, and idle gaps our generator happens to produce. Because the harness itself changed once during this work, we treat its influence as a hypothesis to test rather than assume away. We re-measured the tube/sp–reduction relationship under seven perturbations of the burst generator: burst size scaled down and up (lognormal μ), its dispersion narrowed and widened (lognormal σ), and the inter-burst gaps made heavier and lighter (Pareto α). A positive control confirmed the canonical setting reproduces the in-sample relationship ($r = +0.63$) before any perturbation was trusted.

The direction of the law held in all seven variants, with the correlation ranging only over $[+0.53, +0.79]$ and significant throughout. The variants that give a congestion-aware rule *more* to do—larger bursts ($r = +0.75$) and wider spread ($r = +0.79$)—strengthen the relationship rather than weaken it, exactly as the mechanism predicts: more contention means more of the outcome turns on routing rather than on the fixed structural floor. The law is a property of the topologies, not of the bench on which they were stressed.

13 Discussion

13.1 What the equivalence means

The result of this paper is that two classical-mechanical forces—gradient descent on a potential and a Newton's-third-law reaction to loss—reach the operating point of purpose-built load balancers across the topologies where the geometry leaves room to manoeuvre. We read this less as a statement about routing and more as a statement about the problem: where the topology admits many near-shortest alternatives, good load distribution is not a feat requiring elaborate machinery, but something a simple local physical rule recovers on its own. The engineering and the physics converge because they are both descending the same landscape; the engineering simply pays for an explicit, enumerated map of it that the physics navigates implicitly.

This reframes our own project history. The work began by asking what would happen if a packet were a physical particle with mass, heat, and friction, and accumulated machinery in pursuit of a decisive win. The decisive win never materialised against strong baselines; what materialised instead, once we stopped trying to beat the engineering and started measuring whether we could match it, was a cleaner and more defensible claim. Parsimony was the result, not the consolation.

13.2 Limitations

We state the boundaries of the claim plainly.

- **Simulation, not deployment.** All results are from a flow-level simulator, not a hardware or packet-level testbed. The simulator captures sustained-flow congestion dynamics but abstracts away queueing detail, control- loop latency, and the operational cost of maintaining the scar state in a real forwarding plane. In particular, micro-bursts—the microsecond-scale queue excursions that DRILL's per-packet reactivity is designed to absorb—do not exist at flow granularity; the equivalence reported against DRILL is therefore established for sustained-flow traffic and is not claimed for the micro-burst regime. Testing it requires a packet-level instrument (an ns-3-class simulator or an emulated data plane), outside the scope of the flow-level harness used here.

- **Synthetic demand on real topologies.** An earlier revision of this harness loaded the GÉANT graph under the Abilene label; the audit and the fix are reported in Section 6. Both backbones now run as genuinely independent topologies, but their demand matrices remain synthetic (gravity-style) rather than measured traffic: results on them should be read as topology-faithful, demand-approximate.
- **Predictor, not law.** The tube/sp relationship has $R^2 = 0.60$: a strong band, not a deterministic rule, with scale-free topologies fitting loosest.
- **Margin choice.** The equivalence margin ($\delta = 2$ percentage points) is a modelling decision; we report raw drop rates so the verdicts can be reinterpreted under a different margin.

13.3 Future work: one principle, other domains

If load distribution on a graph is recoverable from two classical forces, the natural question is whether the same two-term rule applies to flow problems that are not networking at all. Last-mile logistics and supply-chain routing share the essential structure—flows over a capacitated graph, congestion as a cost to be spread—and admit public datasets. A companion study applying the identical core to such a domain would test whether what we report here is a networking trick or an instance of something more general: a classical-physics principle for flow allocation on graphs, of which packet routing is the first case. We regard that generalisation as the most interesting direction this work opens, and as the proper test of whether the equivalence reported here reflects a property of physics or merely a property of networks. Two narrower directions emerge from the analysis above but lie beyond, or remain ahead of, the present instrument. An *adaptive scar half-life*: an operator does not know failure timescales in advance, and a slow meta-controller estimating $T_{1/2}$ from the recent variance of losses would make the temporal channel self-tuning. And a *packet-level replication* targeting the micro-burst regime excluded above, ideally on an emulated control plane, which would also price the telemetry this paper's simulator grants for free.

13.4 Conclusion

A ping packet, treated as a particle subject to a gradient and Newton's third law, routes about as well as engineered load balancers on the topologies where the geometry allows—and a single structural ratio says in advance which those are. The machinery that modern load balancers bring is not, on those topologies, buying performance the physics cannot reach; it is buying an explicit path enumeration that the physics performs implicitly. Two terms, and a condition for when they suffice. That is the whole of it.

13.5 The control-plane cost we are not hiding

The reference core fits in about sixty lines of Python, and it is tempting to read that as computational frugality. It is not, and we want to be explicit about the gap, because it is the most consequential limitation of this work.

Two distinct notions of parsimony must be separated. The core is parsimonious in *mechanism*: two terms, no per-flow state, no precomputed path tables, no hand-tuned controller. It is *not* demonstrated to be parsimonious in *computation or control overhead.* Algorithm 1 evaluates a min-potential path by dynamic programming over the hop budget, $O(k|E|)$ per decision, and to evaluate $\phi(u, v)$ on each edge it needs the instantaneous load x and scar state s of every edge in the graph. Our flow-level simulator provides that global state for free and instantaneously. A real deployment would not: the load and loss state of remote edges would have to reach the deciding node through telemetry or flooding, with latency and overhead that our model omits entirely.

Table 10: Stale-state sweep, pooled over 15 topologies $\times$ 6 paired seeds. The router decides on a load view refreshed every T ticks. Oscillation ratio is the mean temporal variance of per-edge load relative to T=1. (`experiments/stale_state.py`, `data/stale_state.json`)

T	mean drop	Δ vs T=1 (pp)	Wilcoxon p	osc. ratio
1	0.028	—	—	1.00
2	0.033	+0.44	9.9×10^{-5}	1.02
5	0.053	+2.49	8.6×10^{-14}	1.10
10	0.082	+5.39	8.0×10^{-16}	1.18
20	0.105	+7.70	1.2×10^{-15}	1.26
40	0.118	+8.94	8.3×10^{-16}	1.29

This matters for the comparison as much as for deployability. DRILL, in particular, is a switch-local, per-hop mechanism that reacts to local queue occupancy and assumes no global view; CONGA distributes congestion state but within a bounded fabric. By granting our core an idealized, zero-latency global state, the simulator may flatter it relative to a genuinely distributed, local-information baseline. How the equivalence degrades when the global state is stale was, until this revision, the single most important experiment this work left open; Section 13.6 now reports it, together with a per-decision cost benchmark and a bursty-arrival stress. We therefore frame the core's standing as *equivalence in routing quality under idealized state visibility*, not as a claim of operational efficiency. Establishing whether the core survives the remaining constraints—telemetry bandwidth and propagation dynamics beyond the staleness and visibility priced in Section 13.6—is still required before any deployability claim could be made.

13.6 Pricing the deployment assumptions: stale state, decision cost, bursty arrivals

An external review round of nine independent readers converged on exactly the two limitations declared above: the idealized global state, and the flow-level granularity. Rather than leave them as confessions, we measured them. Two predictions were registered before running the staleness experiment: ours, that degradation at one-twentieth telemetry frequency would stay under half a percentage point; and a reviewer's, that deciding on stale potential maps would induce resonance—severe oscillation that amplifies congestion instead of relieving it. Both turned out to be wrong, in instructive and opposite ways.

Stale state. The validated core (gradient plus Archimedean threshold) routes every flow using a load *view* that is refreshed only every T ticks, while the true load evolves underneath; T=1 (one-tick lag, the minimum any real system pays) is the control. Fifteen topologies, six paired seeds, $T \in \{1, 2, 5, 10, 20, 40\}$. Table 10 reports the pooled result. The degradation is real and monotone: negligible at half frequency (+0.44pp), material from one-fifth frequency onward (+2.5pp at T=5, +7.7pp at T=20). Our registered prediction missed the magnitude badly, and we report that miss in the same spirit as the other inversions in this paper. The reviewer's predicted failure mode, however, did not appear at all: the mean temporal variance of per-edge load—the natural signature of oscillatory rerouting—rises only from 1.00× to 1.29× across the entire sweep, with no inflection. Stale state makes the core *blind*, not *unstable*: decisions lose information and quality degrades smoothly, but the attractor does not turn into a feedback amplifier. Operationally, the freshness assumption now has a price tag: telemetry at half the decision rate costs under half a point; at one-fifth the decision rate the cost consumes the equivalence margin itself.

Decision cost. On the same substrate (pure Python, identical graphs, warm load state), one routing decision of the physical core costs 72–2,347 μs depending on graph size, the $O(k|E|)$ scaling visible directly. Evaluating CONGA's K=16 precomputed catalogue costs roughly ten times more *in this substrate*, and DRILL's switch-local sampling is 5–26× cheaper. None of these numbers is a hardware

Table 11: Spatial visibility sweep, pooled over 15 topologies $\times$ 6 paired seeds. True load is visible within k hops of the deciding node, zero beyond.

k	mean drop	vs. global (pp)	gap recovered
0 (blind)	0.105	+7.66	0%
1	0.077	+4.87	36%
2	0.056	+2.80	63%
3	0.046	+1.75	77%
global	0.028	—	100%

measurement—all three mechanisms deploy on optimized datapaths—but the comparison within a common substrate supports a bounded claim: the core hides no anomalous decision cost relative to its flow-level rivals, and at 0.07–2.3 ms per *flow* (not per packet) it is feasible in software at backbone flow-arrival rates. Per-packet, microsecond-scale switching remains DRILL-and-silicon territory. (`experiments/cpu_bench.py`)

Bursty arrivals (flow-level). A schedule of calm traffic punctuated by dense five-tick bursts of short flows every twenty-five ticks—the harshest arrival pattern expressible at flow granularity—opens a gap in DRILL's favour of +0.59pp pooled ($p = 0.014$): its territory, confirmed and now quantified at roughly half a point. Thirteen of fifteen topologies remain within the 2pp equivalence margin. Against CONGA-K=16 the core concedes nothing (mean -0.74pp in the core's favour, not significant; the mean is driven largely by Abilene, where the precomputed catalogue pays heavily under bursts, while CONGA wins ten of fifteen topologies by small margins; fourteen of fifteen within 2pp). True microsecond micro-bursts remain out of scope as stated above. (`experiments/bursty_bench.py`, `data/bursty_bench.json`)

Partial visibility (the spatial axis). The complementary question is spatial: instead of a stale global snapshot, the deciding node sees the *true* load only on edges within k hops of itself; beyond that horizon edges show zero load and the potential degenerates to the static gradient. k=0 is the blind router; the blind-to-global gap is 7.7pp pooled. Two checks were registered before running: k=2 recovers at least half the gap, k=3 at least two-thirds. Both held (Table 11): one hop of horizon recovers 36%, two hops 63%, three hops 77%—at which point the core sits +1.75pp from the global view, *inside the equivalence margin itself.* Vision pays steeply and locally: most of the value of congestion awareness lives in the deciding node's neighbourhood, where its flows land first. The physical core does not need a god's-eye view; it needs a neighbourhood. (`experiments/khop_visibility.py`, `data/khop_visibility.json`)

After this battery, what remains unmeasured: telemetry bandwidth and propagation dynamics beyond the staleness and visibility knobs above, and the packet-level regime—a narrower perimeter, still on the queue.

Data and Code Availability

Every experiment script, the raw per-seed JSON files behind every number in this paper, the results manifest mapping each claim to the script and data file that produce it, and the full audit log are publicly available at `github.com/Carloscodix/EMMET` (MIT license).

A The tube/sp ratio: definition and elementary properties

Throughout, $G = (V, E)$ is the undirected network graph and $d(\cdot, \cdot)$ denotes unweighted hop distance.

Definition (near-shortest tube). Fix a budget factor $b \geq 1$. For an ordered pair (s,t) with $d(s,t) = h$, the *tube* is the edge set

$$T_b(s,t) \;=\; \Big\{ \{u,v\} \in E \;:\; \min\big(d(s,u) + 1 + d(v,t),\ d(s,v) + 1 + d(u,t)\big) \leq \lceil b\,h \rceil \Big\},$$

i.e. the edges that some $s \to t$ *walk* of length at most $\lceil b\,h \rceil$ can traverse. The per-pair ratio is $|T_b(s,t)|/h$, and the graph-level metric reported in this paper is the mean of that ratio over up to 400 uniformly sampled pairs, computed with $b = 1.25$ (`experiments/sweep_topologies.py`).

Proposition 1 (lower bound). $|T_b(s,t)| \geq d(s,t)$ for every connected pair, hence tube/sp ≥ 1 on every graph. *Proof.* Every edge $\{u,v\}$ on a shortest $s \to t$ path with u at distance i from s satisfies $d(s,u) + 1 + d(v,t) = i + 1 + (h - i - 1) = h \leq \lceil b\,h \rceil$. A shortest path has h edges. □

Proposition 2 (monotonicity). $T_b(s,t) \subseteq T_{b'}(s,t)$ whenever $b \leq b'$; the metric is nondecreasing in the budget factor. *Proof.* Immediate from the definition. □

Proposition 3 (trees anchor the law exactly). On a tree, every pair (s,t) has a unique simple path, so *any* loop-free routing mechanism—blind or congestion-aware, simple or engineered—produces the identical edge-load distribution: divergence is exactly zero. The lower anchor of the two-factor law is therefore not an empirical observation but a degeneracy of the path space. □

Observation (walks, not simple paths: a mechanical source of residual). T_b counts edges reachable by *walks* within the budget, which is a superset of the edges usable by simple paths. On a tree with $h \geq 1/(b-1) \cdot 2$, an edge hanging one hop off the shortest path enters T_b even though no simple $s \to t$ path can use it; the same overcount occurs around pendant vertices and hub peripheries in general graphs. The computed metric is therefore an *upper bound* on the router's true action support, loosest exactly where low-degree appendages meet high-degree cores—which is the anatomy of scale-free graphs, the family this paper reports as fitting the regression loosest. We state this as the identified mechanical component of the predictor's residual ($R^2 = 0.60$); a corrected simple-path tube is a natural refinement and remains future work.

Connection to the routing core. By construction, $T_b(s,t)$ contains every edge that any router operating under hop budget b can place the (s,t) flow on. All redistribution that any congestion-aware mechanism performs is therefore supported inside the tube, and the per-hop tube width $|T_b(s,t)|/h$ measures the size of the action space per unit of task length. This is the mechanistic reading of the empirical law: the gain of congestion-aware routing over a blind baseline grows with the room the graph offers (Section 10), and the break-even near tube/sp ≈ 2.45 is the bench's estimate of where that room stops paying for the machinery.

Open problem. Derive the functional form of the gain–tube/sp relation and the break-even point from first principles. One candidate is already eliminated: a min-cut account predicted the wrong sign of the structure–divergence relation and was retracted (Section 10). The walk-vs-path correction above is the natural first refinement for any future attempt.

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